

A&E Datamart user guide

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Version 2.4



Translations



Easy read



BSL



Audio



Large print



Braille

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Version history

Version	Date	Summary of changes
1.0	July 2010	Original version.
1.1	Mar 2011	VE55 added, error report location changed and some various minor changes to text
1.2	May 2018	Updates to introduction text and other general updates including screen shots and links. Significant facility added to data items and updates to validation rules. New section added for resubmission of old data (Elizabeth Cole, Data Management).
1.3	Jul 2018	Addition of alt text to images.
2.0	Oct 2020	Updates reflecting new submission processes, file formats and validation for weekly submissions (Elizabeth Cole, Data Management).
2.1	Jan 2021	(Karen Heatlie, Data Management)
2.2	May 2023	(Tara Copic, Data Management)
2.3	Mar 2024	Links updated (Craig Reid, Data Management)
2.4	June 2024	Resubmission Guidance added and advice around submitting a second file. (Craig Reid, Data Management)

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Linked documentation

- [A&E Data Manual](#)
- [A&E Datamart Submissions SOP](#)
- [AE2 File Extract Specification](#)

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1. Introduction

The A&E data mart was established in June 2007 to monitor the compliance of each NHS board against the 4-hour wait standard. Originally, the focus was around a small number of fields such as arrival time, discharge time and reason for wait.

In July 2010 the A&E data mart was extended further to collect items such as diagnosis, several injury fields and an 'alcohol involved' flag, to improve functionality, analysis and reporting.

Significant facility was added in 2016 to enable identification of acute assessment unit and ambulatory care activity.

In 2019, seven new data items were added to record journey time points that are of particular interest to the Scottish Government and also ECDS (Emergency Care Data Set) diagnosis category.

The A&E dataset now contains 63 data items for collection. Some of the data items are mandatory for inclusion, such as the date and time of attendance, and some are optional, such as the injury data items.

There are two types of data submitted to the A&E datamar:

- episode level data
- aggregate level data

Episode level data is now submitted from all emergency departments in Scotland, and a growing number of community A&E departments and minor injury units. Small A&E sites may submit aggregate data which contains only the number of attendances and the number who spent 4 hours or less in the department. This data can only be used to monitor monthly activity and the compliance with the 4 hour standard.

A&E data is submitted to Public Health Scotland (PHS) via SWIFT or AFT to a weekly deadline. A reminder email is sent out at the start of the week advising that

the submission deadline is 5pm every Tuesday, although there is flexibility to allow submissions on Wednesday.

The A&E files are usually for the previous week (e.g. Monday to Sunday) but if a board has issues they have the option to submit two weeks data the following Tuesday.

The Redesign of Urgent Care, December 2020, has requested that boards plan towards submitting daily episode level data – an implementation date has not been set as yet but boards are advised that if they can submit this daily data they should.

Some boards are still unable to submit aggregate data in the new format aggweek file, so the files are submitted monthly in the old format aggregate file. A&E Monthly Files can come in from first of the month and are always for the previous month (for example, January files will come in from 1 February onwards).

A submission reminder email is sent out at the end of each month around the 28th to the 1st advising of the monthly submission deadline.

The boards have a deadline of the 10th of the month to load their files.

The Redesign of Urgent Care, December 2020, has requested that boards plan towards submitting episode level data.

2. File format

Two types of file can be submitted to the A&E data mart:

- **Episode** – a detailed file containing a record for each patient.
- **Aggregate** – a summary file containing a record for each location.

There are currently three types of file that can be submitted to the A&E datamart:

- **Episode** – new format weekly file that contains a detailed record for each attendance.

- **Aggregate** – contains only the total attendances for each location. This is the old format monthly file and will eventually be phased out.
- **Aggweek** – contains only the total attendances for each location, this is the new format weekly file for submitting aggregate data:.

The Redesign of Urgent Care, December 2020, has asked that moving forward, all Sending Locations should submit episode level data in the updated format.

The episode level files can accommodate data submitted for any time frame, i.e. it does not need to contain data for a single month or week. The old format monthly episode file is still available but no longer used by most boards.

Files must be .DAT file format and contain a header record in the first line followed by the data structure. The data structure is a set of records where each record relates to an attendance (episode file) or to a location containing summary attendance figures (aggregate file).

The following file naming convention should be used:

- **AE_FILETYPE_SENDINGLOCATION_CCYYMMDD**

For example:

- AE_AGGREGATE_A103H_20100110
- AE_AGGWEEK_A103H_20100110
- AE_EPISODE_A111H_20100110

The file extension will be '**DAT**'

Where the date in the file name is the date the extract was taken.

The file will fail if the correct naming convention is not used.

2.1. Header record

The header record is the first row of a file and consists of several items that are used to identify the file, allowing the file to be validated before it is loaded into the datamart. The header record must be present in all files, both episode and aggregate level, with each item separated by a comma. The information in the header record must be consistent with the information supplied in the file name.

The header record must contain **all** of the following items.

No.	Name	Format	Description
1	File Reference	Char (3)	Identifies the project and file data type. • AEU–A&E Episode level data updated (new format for weekly submissions) • AEW – A&E Aggregate Week level data (new format for weekly submissions) • AES – A&E Aggregate Month level data (old format) • AEE – A&E Episode level data (old format, no longer used)
2	Sending Location Code	Char (5)	The NHS Location that has submitted the data, i.e. S314H
3	Date of Extract	Char (8)	The date the extract was taken in the format CCYYMMDD (e.g. 20200301 for 1 March 2020).
4	Data Year Month	Char (6)	Year and month of the A&E data in the format CCYYMM, e.g. 202001, for monthly aggregate files. Week the A&E data relate to in the format CCYYMMDD, e.g. 20200112 which reflects the last day of the weekly submission where Sunday is always the last day of the week.
5	Submission Reference Number	Char (6)	A unique number identifying the number of the submission from a certain location. The submission number must increment per

No.	Name	Format	Description
			successful validation for each sending location. Files which are invalid and rejected for any reason must be corrected and resubmitted with the same Submission Reference Number as the original.
6	Record Count	Integer	The number of records in the file.

Notes:

1. Data Year Month will be ignored for processing the episode level files.
2. The file should not contain column name headings.

Examples of a header record:

- **AEU,S314H,20200114,20200112,2,5000**

Episode level data sent from the Royal Infirmary of Edinburgh extracted on 14 January 2020 containing data for the week Monday 6 January to Sunday 12 January 2020. The submission reference number is 2 and there are 5,000 records in the file.

- **AEW,A216H,20200603,201805,5,13**

Aggregate level data sent from Girvan Community Hospital extracted on 18 February 2020 containing data for the week Monday 10 to Sunday 16 February 2020. The submission reference number is 5 and there are 13 records in the file.

- **AES,N151H,20200106,201912,5,13**

Aggregate level data sent from Aboyne Hospital extracted on 6 January 2020 containing data for the month of December 2019. The submission reference number is 5 and there are 13 records in the file.

2.2. Data content format

The tables below give an overview of the data items to be included in the records of the aggregate and episode level files.

Following the DAT file format, all data items must be separated by a comma.

2.2.1. Aggregate level data

No.	Data item	Format	Type	Description
1	Location code	Varchar(5)	Mandatory	The hospital location code.
2	Number of Attendances	Varchar(6)	Mandatory	The number of A&E attendances.
3	Number Meeting Target	Varchar(6)	Mandatory	The number of A&E attendances seen within 4 hours.

Note:

- In aggregate files **only** new and unplanned return attendances should be counted in the attendance figures (planned returns and recalls should be excluded).

2.2.2. Episode file - updated file format

A full list of the codes and definitions for each of the episode level data items is given in the document 'A&E Data Manual'.

No.	Data item	Format	Type
1	Location Code	5 characters	Mandatory
2	CHI Number	10 characters	Optional (but recommended)
3	Case Reference Number	6 to 14 characters	Mandatory

No.	Data item	Format	Type
4	Patient Year of Birth	4 digits	Optional
5	Arrival Mode	2 characters	Mandatory
6	Referral Source	2 to 3 characters	Mandatory
7	Place of Incident	2 to 3 characters	Optional
8	Triage Category	2 digits	Optional
9	Patient Management Type	2 digits	Optional
10	Diagnosis1	2 digits	Optional (but recommended that at least one of 10, 11 or 12 are recorded)
11	Disease1	6-character ICD10 code	Optional (but recommended that at least one of 10, 11 or 12 are recorded)
12	Diagnosis1 Text	100 characters	Optional (but recommended that at least one of 10, 11 or 12 are recorded)
13	Diagnosis2	2 digits	Optional
14	Disease2	6-character ICD10 code	Optional
15	Diagnosis2 Text	100 characters	Optional
16	Diagnosis3	2 digits	Optional
17	Disease3	6-character ICD10 code	Optional
18	Diagnosis3 Text	100 characters	Optional
19	Date and Time of First Full Clinical Assessment	14-character date/time* (CCYYMMDDhhmmss)	Optional

No.	Data item	Format	Type
20	Date and Time of Completion of Treatment	14-character date/time* (CCYYMMDDhhmmss)	Optional
21	Investigation Type1	2 to 3 characters	Optional
22	Investigation Type2	2 to 3 characters	Optional
23	Investigation Type3	2 to 3 characters	Optional
24	Procedure1	2 to 3 characters	Optional
25	Procedure2	2 to 3 characters	Optional
26	Procedure3	2 to 3 characters	Optional
27	Discharge Type	2 to 3 characters	Optional
28	Discharge Destination	2 to 3 characters	Mandatory
29	Referred To 1	2 to 3 characters	Optional
30	Referred To 2	2 to 3 characters	Optional
31	Referred To 3	2 to 3 characters	Optional
32	Arrival Date and Time	14-character date/time* (CCYYMMDDhhmmss)	Mandatory
33	Attendance Category	2 digits	Mandatory
34	Date and Time of Discharge, Admission or Transfer	14-character date/time* (CCYYMMDDhhmmss)	Mandatory
35	Reason for Wait	2 to 3 characters	Mandatory
36	Patient Flow	1 digit	Mandatory
37	Patient Postcode	8-character postcode	Optional
38	Activity when Injured	2 to 3 characters	Optional
39	Alleged Perpetrator of Violence	2 characters	Optional
40	Bodily Location of Injury	2 to 3 characters	Optional

No.	Data item	Format	Type
41	Common Preventive Measures	2 to 3 characters	Optional
42	Date and Time of Event	14-character date/time* (CCYYMMDDhhmmss)	Optional
43	Time Since Injury	2 characters	Optional
44	External Cause of Injury	2 to 3 characters	Optional
45	Intent of Injury	2 to 3 characters	Optional
46	Nature of Injury	2 to 3 characters	Optional
47	Objects/Substances Involved in Producing Injury	2 to 3 characters	Optional
48	Alcohol Involved	2 characters	Optional
49	Healthcare Professional (HCP) Number	5 to 8 characters	Optional
50	Presenting Complaint	100 characters	Optional
51	Impairment	2 characters	Optional
52	Ethnicity	2 to 6 characters	Optional
53	Religion	2 to 6 characters	Optional
54	Sexual Orientation	2 characters	Optional
55	Significant Facility	2 characters	Optional
56	Date & Time of Triage	14-character date/time* (CCYYMMDDhhmmss)	Optional
57	Date & Time of First Diagnostics	14-character date/time* (CCYYMMDDhhmmss)	Optional
58	Date & Time of Decision to Discharge, Admit or Transfer	14-character date/time* (CCYYMMDDhhmmss)	Optional

No.	Data item	Format	Type
59	Date & Time of Treatment Start	14-character date/time* (CCYYMMDDhhmmss)	Optional
60	Wait to Cubicle	3 characters	Optional
61	Wait to Bed Request	3 characters	Optional
62	Wait to Bed Available	3 characters	Optional
63	ECDS Diagnosis Category	Free text	Optional

* Date and time data items should be in the 14-character format CCYYMMDDhhmmss, the time is in 24-hour format and contains seconds e.g. 20180719212300 (19 July 2018 at 9.23pm).

Note:

Unlike aggregate level data, **all** Attendance Categories should be included in episode level data (i.e. new, planned returns, unplanned returns and recalls).

2.2.3. File name

A requirement of the A&E data mart is that the extract file names must be in the in the format shown below.

In addition, the file extension must be DAT. Files that do not meet these specifications will fail validation.

AE_FILETYPE_SENDINGLOCATION_CCYYMMDD

Where:

- FILETYPE is either EPISODE or AGGREGATE or AGGWEEK
- SENDINGLOCATION is the Sending Location Code
- CCYYMMDD is the date the data was extracted

Three examples of a file name are:

AE_EPISODE_A210H_20180501.DAT

Episode level data sent from Ayr Hospital extracted on 1 May 2018.

AE_AGGREGATE_R101H_20180603.DAT

Monthly aggregate level data sent from Balfour Hospital extracted on 3 June 2018.

AE_AGGWEEK_A216H_20200929.DAT

Weekly aggregate level data sent from Girvan Community Hospital extracted on 29 September 2020.

3. File submission

A&E files can be submitted using SWIFT (Submission With Internet File Transfer) or Automated File Transfer (AFT).

Please get in contact if you would like to set up AFT.

SWIFT is a web-based application designed to allow the submission of data files, easily and securely.

3.1. Logging in to SWIFT

To login to your SWIFT account, go to

<https://apps.nhsnss.scot.nhs.uk/swift/loginT.do>

At the login page, enter your Username and Password and click the Login button.

If you do not already have an account or are having problems accessing your account please contact the AE team on phs.aewaiting@phs.scot

Once you are logged into the application you are taken to the homepage. Down the left-hand side of the application is the navigation menu. This is used to go to the different function pages.

To submit a file, select the UPLOAD FILE TO ISD option from the navigation menu and you will be taken to the file upload page as shown below.

To submit a file:

- Select the correct data mart by clicking on the down arrow next to the Application input box. Select Accident and Emergency 2 from the list (if you do

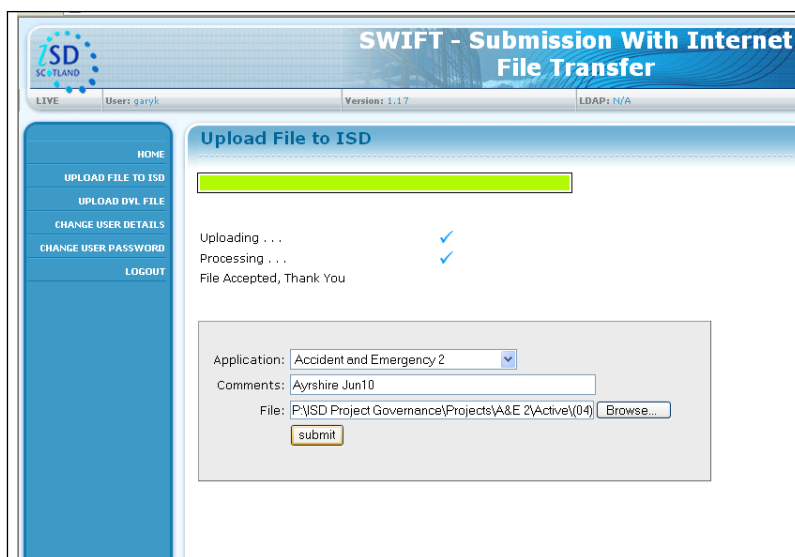
not see this application then you do not have access yet and will need to contact the A&E team).

- Enter a comment in the Comments box; a file will not be accepted without a comment.
- Use the Browse button to locate your file and then click the Submit button.
- Once you have clicked the Submit button you may have to wait a moment for the file to be sent.

A successful upload will be received by a green bar and a File Accepted message, see the screenshot below.

If you do not see this screen then the file has not been sent.

At the same time, you will be sent an email from the SWIFT system to indicate that the file has been sent.



Once a file has been sent you will have to select the UPLOAD FILE TO ISD option from the navigation menu again to submit further files (you can only submit one file at a time).

When you have finished using the application, select the LOGOUT option from the navigation menu.

If you have any problems with this process then please contact the A&E team on phs.aewaiting@phs.scot

3.2. File schedules

Once files are submitted via SWIFT they sit in a staging area before being picked up by the data mart for validation.

A file will only be loaded into the data mart if it passes all of the validation rules.

Files are scheduled to be picked up from the staging area every 30 minutes between 8am and 11pm.

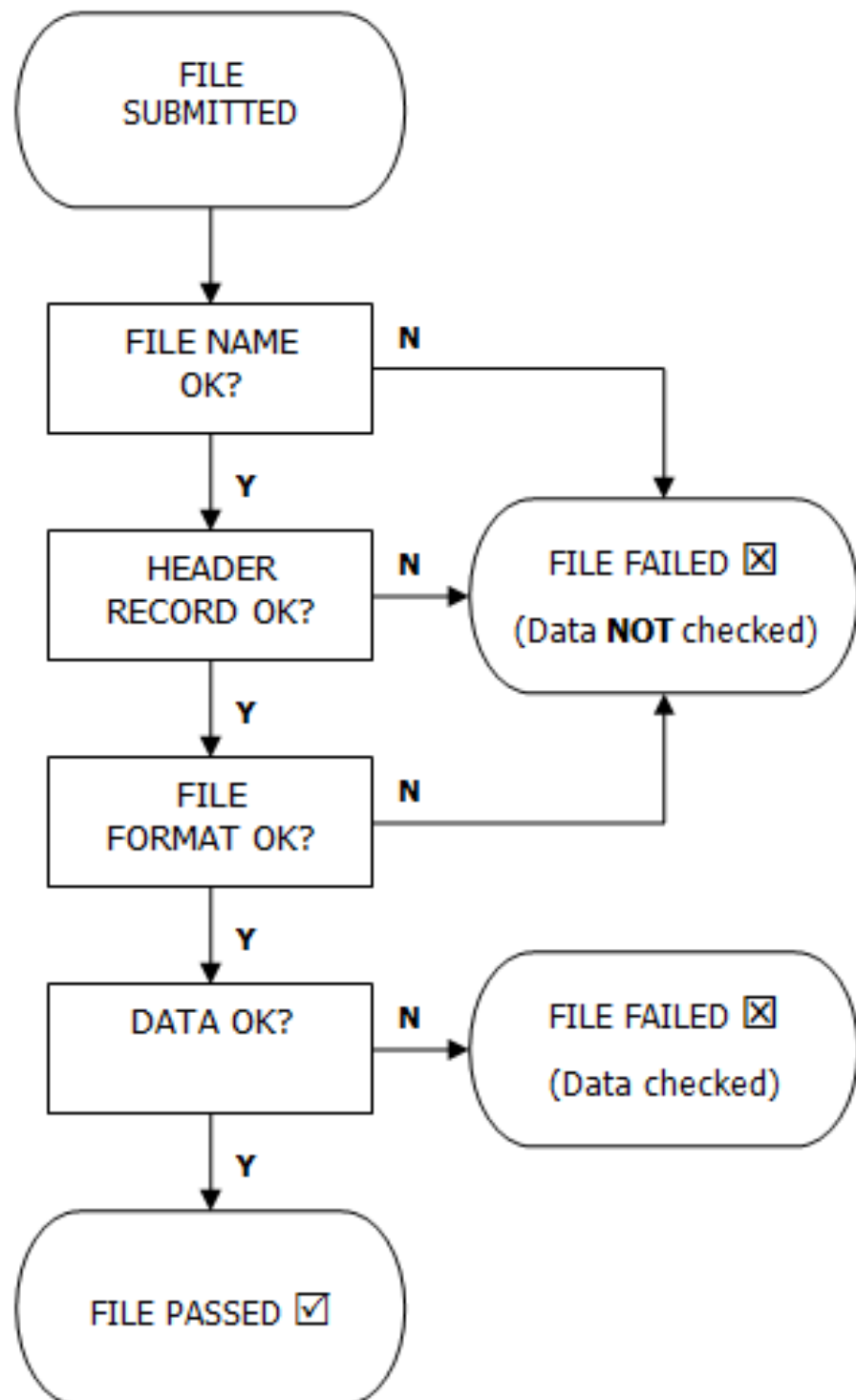
To meet these times, the file will usually need to be submitted around 10 minutes before the pick up time to enable the file to be transferred from SWIFT into the staging area.

It may take much longer to transfer files during busy periods.

4. Validation

Once the file has been submitted to A&E data mart (AE2) via SWIFT it will undergo a number of validation tests to ensure that the file name, header record, file format and data contained within the file match the specification set out in section 2.

Flowchart showing the validation process



Description of process:

1. The file name is first assessed. The sending location code is used to identify the file and is matched to the header record for consistency. If the file name does not meet the specification the file will fail at this point, no more validation tests will be performed and the data within the file will NOT be checked.
2. If the file name is okay then the validation moves on to check the header record. The sending location code must be recognised, the dates must be in the correct format and the record count must match the number of records in the file. If the header record does not meet the specification the file will fail at this point, no more validation tests will be performed and the data within the file will **not** be checked.
3. Once the file passes the file name and header record validation rules the file will then be checked to confirm it is in the correct format. The file is checked to see if it is in .DAT format and that all of the records have the correct number of data items. If the format does not meet the specification the file will fail at this point, no more validation tests will be performed and the data within the file will **not** be checked.
4. If the file passes the first three stages of validation (file name, header record and format) it is only at this point that the actual data in the file will be subject to validation.

A list of the validation rules are given in Appendix A. If any of the validation rules are not met, the whole file will be rejected and the sender will need to correct the error and resubmit the file.

Once the file has been through the validation process you will get an email to notify you if the file has **passed** or **failed** validation.

For a file that has **passed** validation the email will look like this:

AE2 Validation Status



noreply_datawarehouse@nhsnss.scot.nhs.uk

To: phs aawaiting

This sender noreply_datawarehouse@nhsnss.scot.nhs.uk is from outside your organization.

Mail from: </AE2_warehouse_load_jb/AE2_notification_wf/AE2_val_email_processing_wf> sent at: Tue May 2 17:49:59 2023

Message:

=====

The following AE2 source file has PASSED the validation load process:

File Name: ae_aggweek_a216h_20230502.dat

File Identifier: 17044

Data validation details are :

Total Records Processed: 3

Your file is now processing through to the data warehouse, and valid data will be available for reporting via Business Objects in due course.

Please contact "phs.aawaiting@phs.scot" if you have any queries.

For a file that has **failed** validation the email will look like this:

AE2 Validation Status



noreply_datawarehouse@nhsnss.scot.nhs.uk

To: phs aawaiting

This sender noreply_datawarehouse@nhsnss.scot.nhs.uk is from outside your organization.

Mail from: </AE2_warehouse_load_jb/AE2_notification_wf/AE2_val_email_processing_wf> sent at: Tue May 2 17:50:10 2023

Message:

=====

The following AE2 source file has FAILED the validation load process:

File Name: ae_episode_a111h_20230502.dat

File Identifier: 17045

Possible causes include:

- file could not be read/opened
- unrecognised file name and/or file extract type
- one or more header record validation errors
- header control total does not match count of records read
- one or more data records failed validation

Data validation details are :

Total Records Processed: 1141

Filename Error Count: 0

Header Error Count: 0

Data Error Record Count: 1

Data Warning Record Count: 0

Schema Failure: N

Please resolve these issues and resubmit the file.

business objects reports for any errors can be accessed via the link <https://www.bo.scot.nhs.uk/BOE/BI>

Please contact "phs.aawaiting@phs.scot" if you have any queries.

If the file has passed validation it will then be loaded into the data mart.

If the file has failed validation then you will need to take a note of the File ID (File Identifier) number from the email you were sent (7851 in the example above) and obtain the error report as described in the next section.

5. Error reports

To obtain the error report for your failed submission file you will need your Business Objects username and password and the File ID number from the email you were sent.

Go to the following url: <https://www.bo.scot.nhs.uk/BOE/BI>

You should see the following page.

NHS SCOTLAND **NSS BI Reporting**
NHS Scotland

Use of Data in Individual Patient Care or Judging Staff Performance

The information provided to you is derived from systems used in the NHS for the administration of health services or from the registrations held by the General Register Office for Scotland. Although there are quality assurance processes in place, the data may contain undetected inaccuracies about an individual patient, member of staff or department. Therefore the data is not aimed at informing direct clinical decisions about individual patients, or judging the performance of individual staff and should be verified if to be used for either of these purposes.

Data and Statistical Disclosure

The data presented have not been adjusted to protect against potential disclosure risks and are released within this system for management information purposes. The data presented may contain information which enables (perhaps with the aid of further knowledge of the topic) an individual patient or member of staff to be identified. Please ensure access is restricted and that patient confidentiality is not compromised. For further guidance refer to your local NHS Board's disclosure guidelines in the first instance. For ISD this would be ISD's Statistical Disclosure Control Protocol. For SHIS Comparative views, where local guidance does not exist, this would be SHIS SDC Guidance.

Access is Monitored

Please be aware that individual access and usage is monitored and recorded. The data is used to measure system usage and inform of potential areas of improvement. This data may be shared with your own Board but would not be made available outwith your NHS Board.

NHS Scotland is committed to maintaining confidentiality
Every effort has been made to ensure the security of information held by the organisation. If you are authorised to view confidential information which is held by NHS National Services Scotland, you should be aware of the relevant confidentiality requirements and adhere to these requirements.

Audit of access to the Corporate Data Warehouse
Click [here](#) to view details of CDW auditing.
Best Practice - access the Best Practice document [here](#).

Enter your user information, and click "Log On".
If you are unsure of your account information, contact your system administrator.

User Name:

Password:

Authentication:

[Log On](#)

[Help](#)

Check that the authentication is set to LDAP.

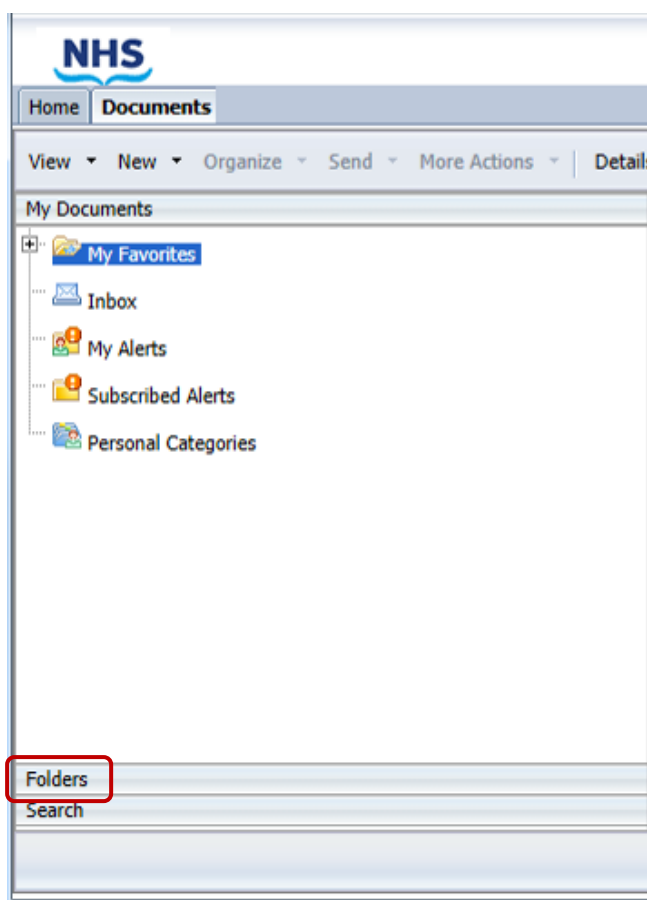
Enter your username and password and select Log On.

If you do not already have an account or are having problems accessing your account please contact the A&E team at phs.aewaiting@phs.scot

After you have logged on, you should be taken to the homepage which shows your recently viewed and run documents.

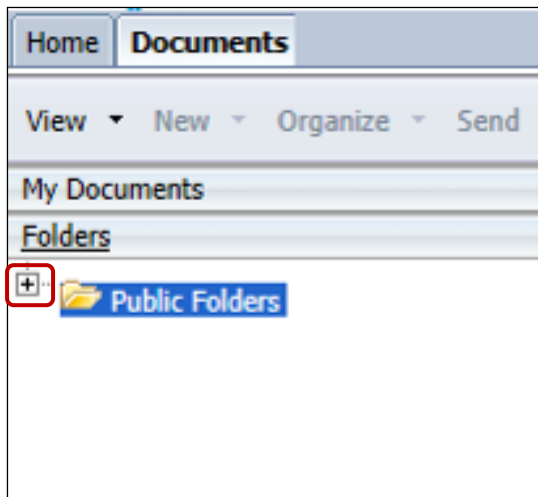
Select the 'Documents' tab on the top lefthand corner, this takes you to a new page which displays folders in 'My Documents'.

This screen is split into two sections – a navigation bar on the left-hand side and a display pane on the right.



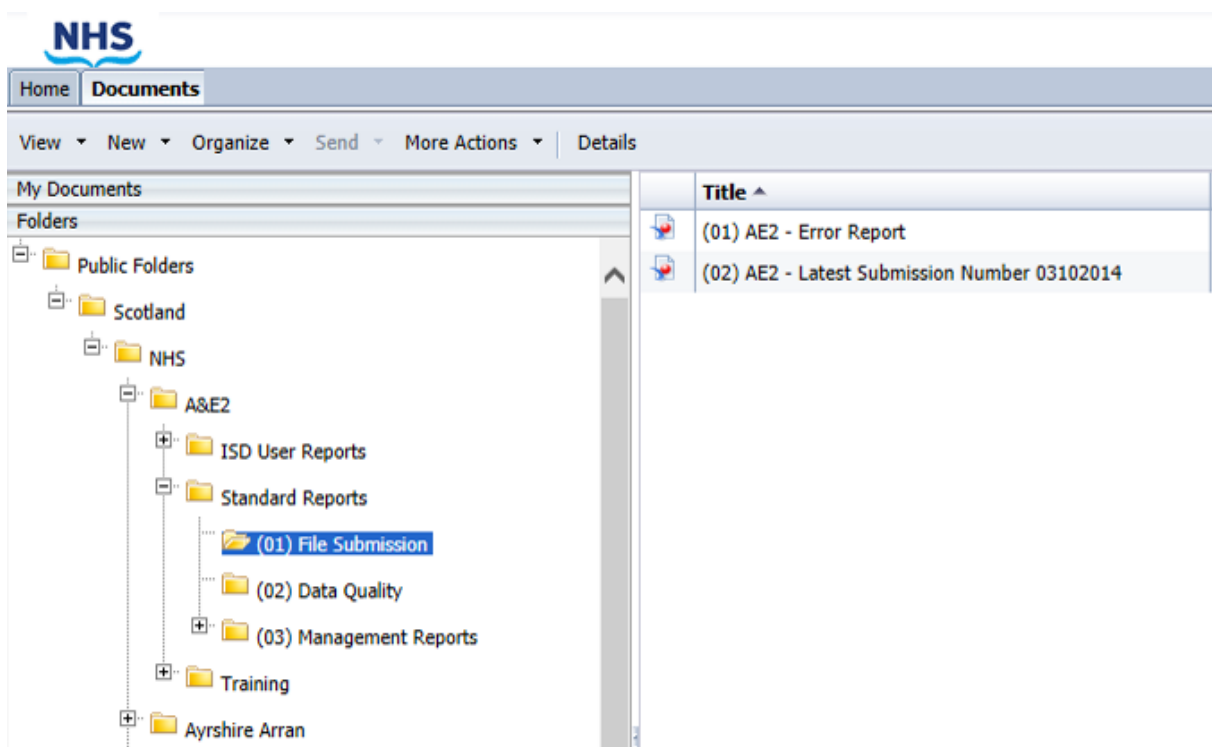
Select the 'Folders' bar at the bottom left of the page.

Navigate through the folders in left-hand pane by clicking the expand button [+] next to 'Public Folder' this will reveal any sub-folders beneath.



To find the error report, go to the following folder in the navigation bar:

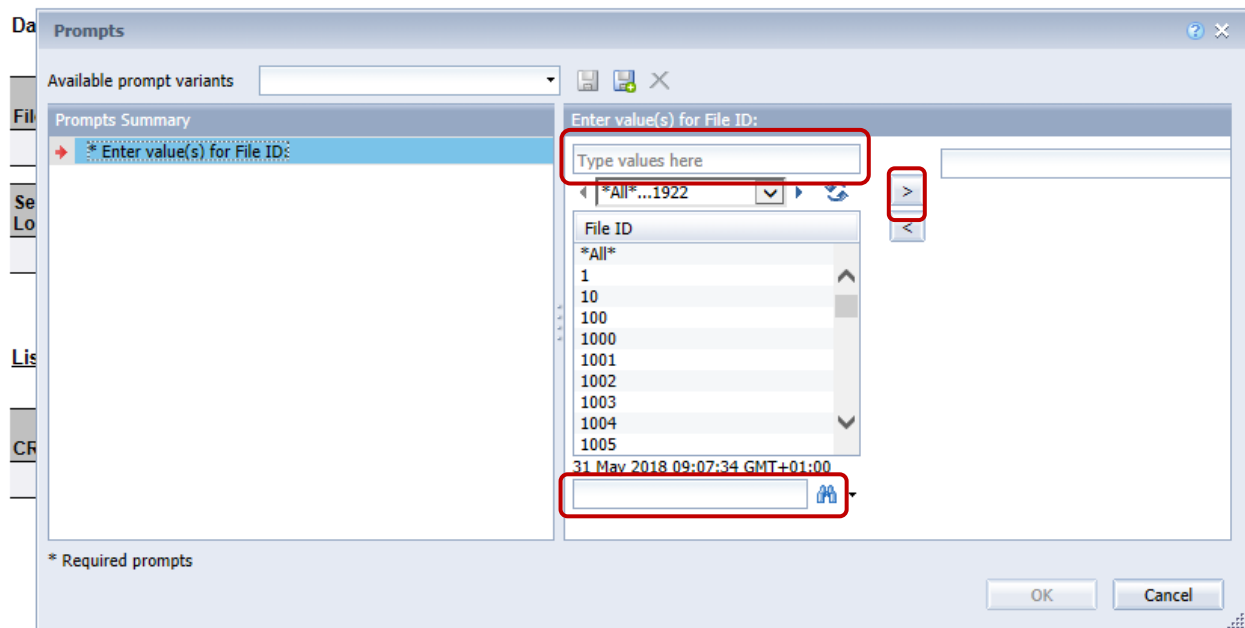
Home/Public Folders/Scotland/NHS/A&E2/Standard Reports/(01) File Submission.



When you click on the '(01) File Submission' folder you should see the files displayed

Find the report named '(01) AE2 - Error Report' in the display pane and click on the report name to open the report, you should be presented with the following screen.

AE2 Data Mart Error Report



You are presented with a Prompt window – this is where you enter the File ID number of the failed file (sent within the failed validation email).

Search in one of two ways.

5. Enter the File ID number in the input box called 'Type values here' and select the righthand arrow button to move the value into right hand box.
6. Enter the File ID in the search box (next to an icon that looks like a small pair of binoculars). The Search button and the File ID should appear in the List window above. Highlight the File ID in the list window and click the arrow button to move the value into right hand box.

If the File ID does not appear, click the Refresh Values button () and repeat the process.

Select the OK button at the bottom righthand side of the prompt window to extract the error report.

You should see a report similar to the following screen.

Home Documents (01) AE2 - Error Report

Web Intelligence | Track | Drill | Filter Bar | Freeze | Outline

Navigation Map

- (01) AE2 - Error Report
 - by CRN
 - by Error Code

AE2 Data Mart Error Report

Date Extracted: 31 May 2018

File ID	Load Date	File Name	File Type
7851	9 May 2018	ae_episode_w106h_20180509.dat	EPISOD

Sending Location Code	Data Month	File Status	Submission Reference	Record Count	No of Errors
W106H	201804	FAILED	96	1	2

List of Errors

CRN	Error Code	Error Message	Value (in square brackets)	No of Errors
	HE2	File Reference must be AEE or AES, and must match the FileType	[AES]	1
	HE4	Extract Date is missing, invalid or out of range. Should be a date in the format yyymmdd and have a value between 01-JAN-2000 and the current date	[20180508]	1

Go to the top lefthand side of the display pane and select the dropdown arrow next to the Export button. Select the option Export Document as and then the type of Excel or PDF file you want.

Web Intelligence

Navigation Map

- (01) AE2 - Error Report
 - by CRN
 - by Error Code

AE2 Data Mart Error Repo

Date Extracted: 31 May 2018

File ID	Load Date	Fi
7851	9 May 2018	ae

Sending

You can then open and save the Error Report as an Excel or PDF file and use it to help identify the errors in the submission file.

The first part of the report gives the file details along with the number of errors. The second part of the report gives a list of the errors in the file. The CRN (Case Reference Number) along with the Error Code and Error Message are given to help identify the particular errors in the file.

A list of the Error Codes can be found in [Appendix 1](#).

If an error message is received – ‘Submission Reference is out of sequence for the supplied Sending Location’ – check that the submission reference number follows on from the last loaded file. The submission reference number only increments if the file loads successfully – it will not increment if a file fails.

If unsure, the correct submission reference number for the sending location can be obtained by contacting PHS.aewaiting@phs.scot or by running the report (02) Latest Submission Number which is saved next to the (01) AE2 - Error Report in Business Objects.

When all the errors are corrected the file can be resubmitted via SWIFT as before. For advice regarding file errors or assistance with running error reports please contact PHS.aewaiting@phs.scot

6. Resubmission guidance

There may be various reasons in which data is needed to be resubmitted to the Accident and Emergency 2 dataset.

- Data quality issue was discovered.
- Wrong data in the datamart.
- Incomplete data in the datamart.
- Recording errors.

Please contact Data Management when you have a resubmission of data, out with normal error reporting on expected files.

Depending on what on which variable requires resubmission, will depend on how a resubmission takes place.

If the information within the record is changing any of the following:

- case reference number
- location code
- arrival date
- arrival time

PHS Data Management firstly need to have the original record deleted, as changing any of the above duplicates the record in the datamart, when it is resubmitted.

If it's a large-scale resubmission that requires deletion and replacing the data, this would require PHS Data Management testing first.

When any of the above variables are being changed, the entire record would need to be resubmitted, not just the variable that is being amended.

If resubmitting data from different data weeks or even years records can go in one resubmission extract with all the fields completed with the corrected data. The file header requires the dates of the day of resubmission, not the dates where the records were originally from.

A resubmission that isn't changing the case reference number, location code, arrival date or arrival time can be done easier as this overwrites the data in the datamart with the corrected data.

For example:

If the arrival mode was being changed from arrival mode (ambulance air) to arrival mode (ambulance road), the full patient record would need to be

resubmitted, not just the arrival mode field, as this is the only field that would populate in the datamart. The entire record needs resubmitted.

When a resubmission is large, over 100 records, data management would need to involve IT to have staging tables and folders backed up. IT may require the data to be uploaded into the UAT environment first; this would depend on the file size and how many fields were being changed. Data Management will be able to advise.

Please contact Data Management phs.aewaiting@phs.scot when you have any kind of resubmission the team can provide further information.

Appendix 1: Validation rules

File reference	Error code	Error type	Error severity	Error message
ALL	FE0	File Name	Error	Invalid File Name: does not match template AE_FILETYPE_SENDINGLOCATION_CCYYMMDD
ALL	FE1	File Name	Error	Invalid File Name: Cannot identify file type. Must be AE_EPISODE, AE_AGGREGATE or AE_AGGWEEK
ALL	FE2	File Name	Error	Invalid File Name: Sending Location is missing or not recognised
ALL	FE3	File Name	Error	Invalid File Name: Date must be in format CCYYMMDD
ALL	FE4	File Name	Error	Invalid File Name: Extension must be .DAT
ALL	HE1	Header	Error	Header record could not be read - check file format
ALL	HE2	Header	Error	File Reference must be AEE, AEU, AEW or AES, and must be consistent with the File Type
ALL	HE3	Header	Error	Sending Location in header record is missing, not recognised or does not match file name.

File reference	Error code	Error type	Error severity	Error message
ALL	HE4	Header	Error	Extract Date is missing, invalid or out of range. Should be a date in the format yyyyymmdd and have a value between 01-JAN-2000 and the current date
ALL	HE5	Header	Error	Data Year Month must be in the format CCYYMM where the File Reference is AES/AEE. Must be in the format CCYYMMDD where the File Reference is AEW/AEU.
ALL	HE6	Header	Error	Data Year Month must be no more than 3 months prior to latest value already loaded (no longer used)
ALL	HE7	Header	Error	Submission Reference is out of sequence for the supplied Sending Location
ALL	HE8	Header	Error	Header Record Count must be numeric
ALL	HE9	Header	Error	Header Record Count (control total) does not match number of data records in file
ALL	SE1	Schema	Error	Data record could not be read - check file format
AES, AEW	VE1	Data	Error	Location Code must be supplied
AES, AEW	VE2	Data	Error	Invalid Location Code
AES, AEW	VE3	Data	Error	Number of Attendances must be supplied
AES, AEW	VE4	Data	Error	Number Meeting Target must be supplied
AES, AEW	VE5	Data	Error	Number Meeting Target must be less than or equal to the Number of Attendances

File reference	Error code	Error type	Error severity	Error message
AEE, AEU	VE6	Data	Error	Location Code must be supplied
AEE, AEU	VE7	Data	Error	Invalid Location Code
AEE, AEU	VE8	Data	Error	Case Reference Number must be supplied
AEE, AEU	VE9	Data	Error	Case Reference Number must be between 6 and 14 characters long
AEE, AEU	VE10	Data	Error	Invalid Arrival Mode
AEE, AEU	VE11	Data	Error	Invalid Referral Source
AEE, AEU	VE12	Data	Error	Invalid Place of Incident
AEE, AEU	VE13	Data	Error	Invalid Triage Category
AEE, AEU	VE14	Data	Error	Invalid Patient Management Type
AEE, AEU	VE15	Data	Error	Invalid Diagnosis1
AEE, AEU	VE16	Data	Error	Invalid Diagnosis2
AEE, AEU	VE17	Data	Error	Invalid Diagnosis3
AEE, AEU	VE18	Data	Error	Invalid Disease1
AEE, AEU	VE19	Data	Error	Invalid Disease2
AEE, AEU	VE20	Data	Error	Invalid Disease3

File reference	Error code	Error type	Error severity	Error message
AEE, AEU	VE21	Data	Error	Invalid Date & Time of First Full Clinical Assessment. Must be in format CCYYMMDDhhmmss
AEE, AEU	VE22	Data	Error	Date and Time of First Full Clinical Assessment must be on or after the Arrival Date and Time
AEE, AEU	VE23	Data	Error	Invalid Date & Time of Completion of Treatment. Must be in format CCYYMMDDhhmmss.
AEE, AEU	VE24	Data	Error	Date and Time of Completion of Treatment must be on or after the Arrival Date and Time
AEE, AEU	VE25	Data	Error	Date and Time of Completion of Treatment must be on or after the Date and Time of First Full Clinical Assessment
AEE, AEU	VE26	Data	Error	Invalid Investigation Type1
AEE, AEU	VE27	Data	Error	Invalid Investigation Type2
AEE, AEU	VE28	Data	Error	Invalid Investigation Type3
AEE, AEU	VE29	Data	Error	Invalid Procedure1
AEE, AEU	VE30	Data	Error	Invalid Procedure2
AEE, AEU	VE31	Data	Error	Invalid Procedure3
AEE, AEU	VE32	Data	Error	Invalid Discharge Type

File reference	Error code	Error type	Error severity	Error message
AEE, AEU	VE33	Data	Error	Discharge Destination must be supplied
AEE, AEU	VE34	Data	Error	Invalid Discharge Destination
AEE, AEU	VE35	Data	Error	Invalid Referred To 1
AEE, AEU	VE36	Data	Error	Invalid Referred To 2
AEE, AEU	VE37	Data	Error	Invalid Referred To 3
AEE, AEU	VE38	Data	Error	Arrival Date and Time must be supplied
AEE, AEU	VE39	Data	Error	Invalid Arrival Date and Time. Must be in format CCYYMMDDhhmmss
AEE, AEU	VE40	Data	Error	Arrival Date and Time must be within the submission month contained in the header record (no longer used)
AEE, AEU	VE41	Data	Error	Attendance Category must be supplied
AEE, AEU	VE42	Data	Error	Invalid Attendance Category
AEE, AEU	VE43	Data	Error	Date and Time of Discharge, Admission or Transfer must be supplied
AEE, AEU	VE44	Data	Error	Invalid Date and Time of Discharge, Admission or Transfer. Must be in format CCYYMMDDhhmmss
AEE, AEU	VE45	Data	Error	Date and Time of Discharge, Admission or Transfer must be on or after the Arrival Date and Time

File reference	Error code	Error type	Error severity	Error message
AEE, AEU	VE46	Data	Error	Date and Time of Discharge, Admission or Transfer must be on or after the First Full Clinical Assessment
AEE, AEU	VE47	Data	Error	Date and Time of Discharge, Admission or Transfer must be on or after the Date and Time of Completion of Treatment
AEE, AEU	VE48	Data	Error	Reason for Wait must be supplied.
AEE, AEU	VE49	Data	Error	Invalid Reason for Wait
AEE, AEU	VE50	Data	Error	The Reason for Wait is not set to '00 - No delay' although the Wait is less than or equal to 4 hours
AEE, AEU	VE51	Data	Error	The Reason for Wait is '00 - No delay' but the Wait is greater than 4 hours
AEE, AEU	VE52	Data	Error	Patient Flow must be supplied
AEE, AEU	VE53	Data	Error	Invalid Patient Flow
AEE, AEU	VE54	Data	Error	Invalid Date and Time of Event. Must be in format CCYYMMDDhhmmss
AEE, AEU	VE55	Data	Error	Invalid Ethnicity
AEE, AEU	VE56	Data	Warning	CHI is invalid (length must be 9 or 10 characters)
AEE, AEU	VE57	Data	Warning	CHI number left-padded with 0 but invalid (value not in warehouse)
AEE, AEU	VE58	Data	Warning	CHI number left-padded with 0, value valid
AEE, AEU	VE59	Data	Error	Arrival Mode must be supplied

File reference	Error code	Error type	Error severity	Error message
AEE, AEU	VE60	Data	Error	Referral Source must be supplied
AEE, AEU	VE61	Data	Error	Significant Facility must be supplied
AEE, AEU	VE62	Data	Error	Invalid Significant Facility
AEE, AEU	VE63	Data	Warning	Duplicate Record Removed: The combination of Location Code, Case Record Number and Arrival Date and Time must be unique within the file.
AEE, AEU	VE64	Data	Warning	Duplicate record selected for inclusion in the data mart.
AEU	VE65	Data	Error	Invalid Date and Time of First Diagnostics. Must be in format CCYYMMDDhhmmss
AEU	VE66	Data	Error	Invalid Date and Time of Decision to Discharge, Admit or Transfer. Must be in format CCYYMMDDhhmmss
AEU	VE67	Data	Error	Invalid Date and Time of Treatment ^{Start} . Must be in format CCYYMMDDhhmmss
AEU	VE68	Data	Error	Invalid Date and Time of Triage. Must be in format CCYYMMDDhhmmss
AEU	VE69	Data	Error	Invalid Wait to Cubicle. Must be numeric.
AEU	VE70	Data	Error	Invalid Wait to Bed Request. Must be numeric.
AEU	VE71	Data	Error	Invalid Wait to Bed Available. Must be numeric.
AEE, AEU	VE72	Data	Error	Duplicate record(s) found within submitted batch of files.
AEE	VE73	Data	Error	AEU data items have been supplied in an AEE file.

Footnote:

The File Extract Type ALL refers to all file types.

The file will only be rejected if it fails a rule where the error severity is 'Error'. The sending location will be expected to correct the error and resubmit the file.